

Brad Cai

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Projects

PBR in Vulkan — C++, Vulkan, GLSL

- Implemented physically based lighting in Vulkan featuring image-based lighting, prefiltered radiance and irradiance environment maps, and BRDF LUT generation for real-time rendering of PBR authored glTF assets with metallic-roughness maps.

Vulkan Engine — C++, Vulkan

- Created a graphics engine in Vulkan using C++ with explicit GPU resource management in the material and global pipelines, descriptor sets, and data buffers along with glTF file loading and a scene and node system supporting real-time rendering with validation and debugging support.

Snow Shader — Godot, GLSL

- Implemented a real-time ground snow deformation effect using camera depth values captured in a render texture, combining fragment and compute shaders with parallax mapping to simulate pressed ground terrain.

Work Experience

Software Engineer, ServiceNow

San Diego, California — August 2019 to January 2025

- Worked on multiple customer facing products as a part of ServiceNow's core platform team.
- Added features and fixes to the back end code of the multi-node server using Java, SQL, MariaDB, syntax parse trees, and multithreaded programming.
- Handled front end development work using Javascript, AngularJS, jQuery, CSS, and HTML.
- Lead development and created software architecture plans that were approved by the internal architecture review board.
- Generated performance data and fixed performance issues using Java Flight Recorder and the IntelliJ profiler.

Education

Bachelor of Science in Computer Science

UC Riverside - Bourns College of Engineering

Riverside, California — September 2015 to June 2019

- Graduated cum laude with a GPA of 3.63.
- Taken courses in Computer Graphics, GPU Programming, Virtual Reality, Graphics and Electronic Games, Concurrent Programming, Intermediate Algorithms, Operating Systems, Data Structures, Software Design, Compiler Design, and Machine Learning.

Skills

- Graphics APIs:** Vulkan, DirectX 11 (D3D11)
- Rendering:** Real-time rendering, graphics pipeline architecture, GPU architecture
- Languages:** C++, C, C#, Python
- Shaders:** GLSL, HLSL
- Debugging:** GPU/CPU profiling, RenderDoc
- Engines:** Godot, Unity
- Math:** Linear algebra, vector math, trigonometry
- Platforms & Tools:** GNU/Linux, Git, command line tools